BRIEF-002: Petabyte-Scale Telemetry Fusion at Industrial Fleets

High-Throughput Streaming & Asset Reliability 2025

STAGE-X.SPACE TECHNICAL ARCHITECTURE BRIEF
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EXECUTIVE SUMMARY:
Streaming vibrational and thermal waveforms into OSIsoft PI and SCADA historians across 1,000+ rotating turbine and compressor assets without polling degradation.

KEY ARCHITECTURE PROOFS:
1. Ingestion: Direct memory-mapped ring buffers eliminate polling-based query pressure.
2. Protocol Support: OPC-UA 1.04 binary protocol with fallback to raw Modbus/TCP.
3. Memory Safety: In-place frame scoring using pooled pre-allocated buffers [zero heap allocs].
4. Store-and-Forward: 72-hour offline buffer capacity for leased-line outage resilience.

FIELD MEASUREMENTS:
- Advance Warning Lead Time: 48 to 72 hours on mechanical failure modes
- Historian Read Pressure: Reduced by 88% compared to legacy polling shims
- Uptime Recorded: 99.2% availability across marine and plant deployments